



NORTH SYDNEY BOYS HIGH SCHOOL

2015 YEAR 12 ASSESSMENT TASK 1

Mathematics Extension 1

General Instructions

- Working time – 55 minutes
- **Reading time – 5 minutes**
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

Class Teacher: Please colour

- ☐ Mr Ireland
- ☐ Mr Lam
- ☐ Mr Lin
- ☐ Ms Ziazaris
- ☐ Mr Berry
- ☐ Mr Zuber

Student Number: _____

(To be used by the exam markers only.)

Question No	1-5	6	7	8	9	10	11	Total	Total
Mark	$\frac{5}{5}$	$\frac{8}{8}$	$\frac{6}{6}$	$\frac{6}{6}$	$\frac{6}{6}$	$\frac{10}{10}$	$\frac{5}{6}$	$\frac{47}{47}$	$\frac{100}{100}$

HSC 2015 ASSESSMENT TASK 1 – MATHEMATICS EXTENSION 1

QUESTIONS 1 -5 are to be answered on the multiple choice answer sheet provided.

QUESTION 1

$$\sum_{r=1}^{25} 2r - 14 =$$

- | | |
|---------|-----------|
| (A) 300 | (C) 462.5 |
| (B) 288 | (D) 450 |

QUESTION 2

The remainder when $x^4 - 3x^2 + 4x + 1$ is divided by $x - 2$ is

- | | |
|--------|--------|
| (A) -3 | (C) 2 |
| (B) 1 | (D) 13 |

QUESTION 3

Which of the following is true for all values of a ?

- (A) $\int_{-a}^a 2x^4 dx = 0$
- (B) $\int_{-a}^a 2x^3 dx = 0$
- (C) $\int_{-a}^a (x^2 - 1) dx = 0$
- (D) $\int_{-a}^a |x| dx = 0$

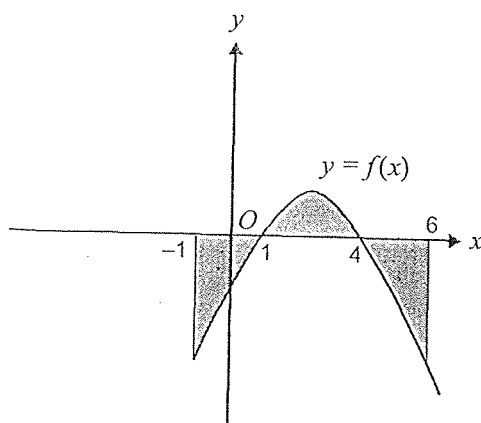
QUESTION 4

Which of the following is TRUE?

- (A) The zero polynomial does not have an infinite number of zeros.
- (B) If a quartic has two distinct real zeros, then they must be double zeros.
- (C) If a quartic polynomial has 4 distinct real zeros then it can be factorised into four real linear factors.
- (D) $\sqrt{7}x - 5$ is not a polynomial.

QUESTION 5

$$\int_{-1}^6 f(x) dx =$$



(A) $-\int_{-1}^0 f(x) dx + \int_0^6 f(x) dx$

(B) $\int_1^4 f(x) dx + 2 \int_4^6 f(x) dx$

(C) $-\int_{-1}^1 f(x) dx + \int_1^4 f(x) dx - \int_4^6 f(x) dx$

(D) $-\int_{-1}^1 f(x) dx + \int_1^4 f(x) dx - \int_4^6 f(x) dx$

QUESTIONS 6 – 11 are to be answered in the answer booklet provided.

QUESTION 6 (Start a NEW page)

- (a) Find the linear factors of $x^3 + 7x^2 + 14x + 8$ 3
- (b) The polynomial $P(x) = x^3 - 4x^2 + kx + 12$ has zeros α, β, γ .
- Find
- (i) $\alpha + \beta + \gamma$ 1
 - (ii) $\alpha\beta\gamma$ 1
 - (iii) the third zero given that two of the zeros are equal in magnitude but opposite in sign. 1
 - (iv) the value of k . 2

QUESTION 7 (Start a NEW page)

(a) Sketch $y = x^3(x+1)(2-x)^2$ 3

(b) Let $P(x) = (x-1)(x+1)Q(x) + ax + b$

where $Q(x)$ is a polynomial and $R(x) = ax + b$ is the remainder.

Given that $(x+1)$ is a factor of $P(x)$ and when

$P(x)$ is divided by $(x-1)$, the remainder is 6, find the remainder when

$P(x)$ is divided by $(x-1)(x+1)$. 3

QUESTION 8 (Start a NEW page)

(a) $\int \left(1 + \frac{1}{2}x\right)^5 dx$ 2

(b) $\int \frac{du}{u\sqrt{u}}$ 2

(c) A train moving between two points in a straight line records the following velocities in m/s at intervals of 10s.

t	0	10	20	30	40
v	0	15	39	45	45

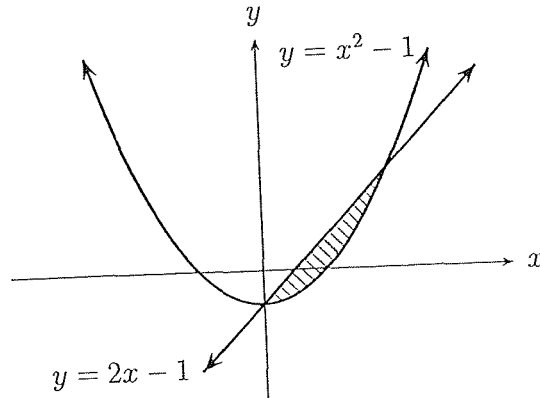
Use Simpson's Rule and five function values

to approximate $\int_0^{40} v dt$. 2

QUESTION 9 (Start a NEW page)

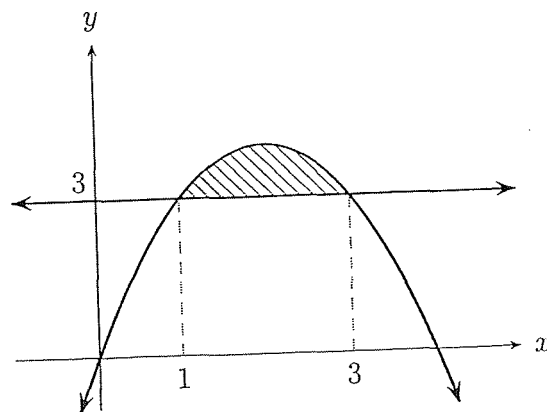
- (a) Find the area enclosed by $y = x^2 - 1$ and $y = 2x - 1$.

3



- (b) Find the volume of the solid obtained by rotating about the X-axis the region enclosed by $y = 4x - x^2$ and the line $y=3$.

3



QUESTION 10 (Start a NEW page)

(a) Find the sum of the geometric series $\frac{1}{2} - 1 + 2 - 4 \dots$ to 15 terms. 2

(b) If $S_n = 3^n + 3n$, find the twelfth term. 2

(c) Find the first term which is less than 0.01 for the geometric series
 $333 + 111 + \frac{111}{3} + \dots$ 3

(d) (i) Show that the series below is geometric.

$$(x) + (1) + \left(\frac{1}{x}\right) + \left(\frac{1}{x^2}\right) + \dots$$
 1

(ii) Find the values of x for which the series above is to have a sum to infinity. 2

QUESTION 11 (Start a NEW page)

(a) (i) Use the remainder theorem to find one factor of
 $P(x) = x(x + m) - n(n + m)$ 1

(ii) By division or otherwise, find the other factor. 2

(b) (i) Show that the derivative of $x^{m+1}(1-x)^n$ is equal to
 $(m+1)x^m(1-x)^n - nx^{m+1}(1-x)^{n-1}$ 1

(ii) Deduce that $\int_0^1 x^m(1-x)^n dx = \frac{n}{m+1} \int_0^1 x^{m+1}(1-x)^{n-1} dx$ 2

END OF EXAMINATON

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

2015 HSC ASSESS TSK 1 - EXT 1 SOLUTIONS

① A ② D ③ B ④ C ⑤ D

6.(a) $x^3 + 7x^2 + 14x + 8$

Test using factor theorem:

$$P(-1) = -1 + 7 - 14 + 8 = 0$$

$\therefore x+1$ is a factor. ✓

$$\begin{array}{r} x^2 + 6x + 8 \\ x+1 \overline{) x^3 + 7x^2 + 14x + 8} \\ \underline{x^3 + x^2} \\ 6x^2 + 14x \\ \underline{6x^2 + 6x} \\ 8x + 8 \\ \underline{8x + 8} \\ 0 \end{array}$$

(or equivalent working) ✓

$$\begin{aligned} \therefore P(x) &= (x+1)(x^2 + 6x + 8) \\ &= (x+1)(x+2)(x+4). \quad \# \end{aligned}$$

(b) $P(x) = x^3 - 4x^2 + kx + 12$ has zeros α, β, γ .

(i) $\alpha + \beta + \gamma = \frac{-(-4)}{1} = 4$ ✓

(ii) $\alpha\beta\gamma = -12$ ✓

(iii) Let zeros be $\alpha, -\alpha, \beta$

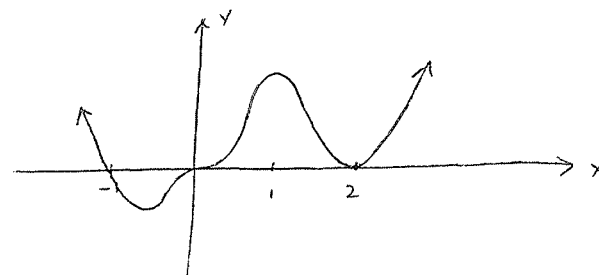
$$\therefore \alpha + (-\alpha) + \beta = 4 \quad \therefore \beta = 4$$

i.e. 3rd zero is 4 ✓

(iv) If one zero is 4, then

$$\begin{aligned} 4^3 - 4(4^2) + 4k + 12 &= 0 \quad \checkmark \\ 4k &= -12 \\ k &= -3 \quad \checkmark \end{aligned}$$

⑦ a) $y = x^3(x+1)(2-x)^2$



b) $P(x) = (x-1)(x+1) \cdot Q(x) + ax + b$

$$P(-1) = 0$$

$$P(1) = 6$$

$$0 = -a + b \quad \text{--- (1)}$$

$$6 = a + b \quad \text{--- (2)}$$

$$2b = 6$$

$$b = 3$$

$$\therefore a = 3$$

$$\therefore \text{Remainder} = 3x + 3$$

⑧ a) $\int (1 + \frac{1}{2}x)^5 dx$

$$= \frac{(1 + \frac{1}{2}x)^6}{6 \cdot \frac{1}{2}} + C$$

$$= \frac{(1 + \frac{1}{2}x)^6}{3} + C$$

b) $\int u^{-3/2} du$

$$= \frac{u^{-1/2}}{-\frac{1}{2}} + C$$

$$= -\frac{2}{\sqrt{u}} + C$$

$$\begin{aligned}
 c) \int_0^{40} v dt &= \frac{10}{3} \left\{ f(0) + 4[f(10) + f(30)] + 2f(20) + f(40) \right\} \\
 &= \frac{10}{3} \left\{ 0 + 4(15 + 45) + 2 \times 39 + 45 \right\} \\
 &= 1210 \text{ m}
 \end{aligned}$$

9. a) Points of Intersection

$$x^2 - 1 = 2x - 1$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x = 0, 2$$

$$A = \int_0^2 (2x-1) - (x^2-1) dx$$

$$= \int_0^2 (2x - x^2) dx$$

$$= \left[x^2 - \frac{x^3}{3} \right]_0^2$$

$$= 4 - \frac{8}{3}$$

$$= \frac{4}{3} \text{ sq. u.}$$

$$b) V = \pi \int_1^3 (4x - x^2)^2 dx - \pi \int_1^3 3^2 dx$$

$$= \pi \int_1^3 (16x^2 - 8x^3 + x^4) dx - \pi \int_1^3 9 dx$$

$$= \pi \left[\frac{16x^3}{3} - 2x^4 + \frac{x^5}{5} \right]_1^3 - \pi [9x]_1^3$$

$$= \pi \left[16 \times 9 - 2 \times 81 + \frac{3^5}{5} - \left(\frac{16}{3} - 2 + \frac{1}{5} \right) \right] - \pi [27 - 9]$$

$$= \frac{136\pi}{15} \text{ c.u.}$$

$$\begin{aligned}
 (10) a) S_{15} &= \frac{a(r^n - 1)}{r - 1} \\
 &= \frac{\frac{1}{2}(-2^{15} - 1)}{-3} \\
 &= 5461.5
 \end{aligned}$$

$$\begin{aligned}
 b) T_{12} &= S_{12} - S_{11} \\
 &= (3^{12} + 36) - (3^{11} + 33) \\
 &= 354297
 \end{aligned}$$

$$c) T_n < 0.01$$

$$ar^{n-1} < 0.01$$

$$333 \left(\frac{1}{3} \right)^{n-1} < 0.01$$

$$\frac{1}{3}^{n-1} < \frac{0.01}{333}$$

$$(n-1) \log \frac{1}{3} < \log \left(\frac{0.01}{333} \right)$$

$$n-1 < \frac{\log \left(\frac{0.01}{333} \right)}{\log \frac{1}{3}}$$

$$n < \frac{\log \left(\frac{0.01}{333} \right)}{\log \frac{1}{3}} + 1$$

$$n > 10.47$$

$$\therefore n = 11$$

$$\begin{aligned}
 \therefore T_{11} &= 333 \left(\frac{1}{3} \right)^{10} \\
 &= 0.0056
 \end{aligned}$$

d) i) If geometric

$$\frac{T_2}{T_1} = \frac{T_3}{T_2}$$

$$LHS = \frac{1}{x}$$

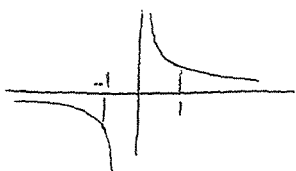
$$RHS = \frac{\frac{1}{x}}{1}$$

$$= \frac{1}{x}$$

$$LHS = RHS$$

(ii) Limiting Sum exists when

$$-1 < \frac{1}{x} < 1$$



True for $x > 1, x < -1$

OR algebraically

$$\frac{1}{x} < 1 \quad \text{and} \quad \frac{1}{x} > -1$$

$$x < x^2$$

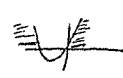
$$x > -x^2$$

$$x - x^2 < 0$$

$$x + x^2 > 0$$

$$x(1-x) < 0$$

$$x(1+x) > 0$$



$x > 1, x < -1$

b) (i) Using Product Rule,

$$\begin{aligned} \frac{d}{dx} x^{m+1} (1-x)^n &= (1-x)^n \cdot (m+1)x^m + x^{m+1} \cdot n(1-x)^{n-1}(-1) \\ &= (m+1)x^m(1-x)^n - nx^{m+1}(1-x)^{n-1} \end{aligned}$$

(ii) From part (i)

$$\int [(m+1)x^m(1-x)^n - nx^{m+1}(1-x)^{n-1}] dx = x^{m+1}(1-x)^n$$

$$\text{i.e.} \quad \int (m+1)x^m(1-x)^n dx - \int nx^{m+1}(1-x)^{n-1} dx = x^{m+1}(1-x)^n$$

$$\therefore (m+1) \int_0^1 x^m(1-x)^n dx - n \int_0^1 x^{m+1}(1-x)^{n-1} dx = [x^{m+1}(1-x)^n]_0^1$$

$$\begin{aligned} (m+1) \int_0^1 x^m(1-x)^n dx &= n \int_0^1 x^{m+1}(1-x)^{n-1} dx + [x^{m+1}(1-x)^n]_0^1 \\ &= n \int_0^1 x^{m+1}(1-x)^{n-1} dx + [1(1-1)^n - 0] \\ &= n \int_0^1 x^{m+1}(1-x)^{n-1} dx + 0 \end{aligned}$$

$$\therefore \int_0^1 x^m(1-x)^n dx = \frac{n}{m+1} \int_0^1 x^{m+1}(1-x)^{n-1} dx$$

(11) a) (i) $P(n) = n(n+m) - n(n+m)$

$$= n^2 + mn - n^2 - mn$$

$$= 0$$

$\therefore x-n$ is a factor.

$$\begin{array}{r} \text{(ii)} \quad x-n \overline{) \begin{array}{r} x^2 + mx + n \\ x^2 + mx - n^2 - mn \\ \hline (m+n)x - n^2 - mn \\ (m+n)x - n^2 - mn \\ \hline 0 \end{array}} \end{array}$$

$$\therefore P(x) = (x-n)(x+m+n)$$

OR $x^2 + mx - n^2 - mn$

$$= (x-n)(x+n) + m(x-n)$$

$$= (x-n)(x+n+m)$$